

SUPPLEMENT ARTICLE

Understanding innate immunity and inflammation in acne: implications for management

B. Dreno,^{1,*} H.P.M. Gollnick,² S. Kang,³ D. Thiboutot,⁴ V. Bettoli,⁵ V. Torres,⁶ J. Leyden⁷ on behalf of the Global Alliance to Improve Outcomes in Acne

¹Department of Dermato Cancerology, University of Nantes, Nantes, France

²Department of Dermatology and Venereology, Medical Faculty, Otto-Von-Guericke-Universitat, Magdeburg, Germany

³Department of Dermatology, Johns Hopkins Medicine, Baltimore, MD,

⁴Department of Dermatology, Pennsylvania State University College of Medicine, Hershey, PA, USA

⁵Department of Dermatology, Arcispedale S. Anna, University of Ferrara, Ferrara, Italy

⁶Department of Dermatology, Juarez Hospital, Mexico City, Mexico

⁷Department of Dermatology, University of Pennsylvania School of Medicine, Philadelphia, PA, USA

*Correspondence: B. Dreno. E-mail: brigitte.dreno@wanadoo.fr

Abstract

Acne has long been understood to have a complex physiological basis involving several main factors: hormonally-stimulated sebum production, abnormal keratinization of the pilosebaceous duct, and an inflammatory immune response to *Propionibacterium acnes*. Recent studies at the molecular and cellular level have begun clarifying how all of these factors interact, and the role of the innate immune system is better appreciated. Inflammation has been demonstrated in all acne lesions - the preclinical microcomedo, comedones, inflammatory lesions, 'post-inflammatory' erythema or hyperpigmentation, and scarring. Inflammation localized to the pilosebaceous unit can be considered the defining feature of acne and should be addressed via multiple therapeutic pathways. Clinicians tend to think oral antibiotics should be used to 'calm' inflammatory acne, but there is good evidence showing that topical retinoids also have anti-inflammatory properties as a class effect. For best therapeutic outcomes, most patients with acne should be treated first line with a topical retinoid plus an antimicrobial agent, as has been demonstrated in thousands of patients involved in clinical trials and recommended by the Global Alliance to Improve Outcomes in Acne for more than a decade. Moving away from reliance on antibiotic therapy for acne is particularly important in an era of worsening antimicrobial resistance and worldwide calls to reduce antibiotic use. Improved understanding about the role of *P. acnes* and the innate immune system in acne should help clinicians in designing efficacious treatment strategies.

Received: 14 April 2015; Accepted: 14 April 2015

Disclosures

The author declares no conflict of interest

Inflammation in acne: focus on activation of innate immunity and the follicular environment

The current understanding of acne is that it has a complex, multifactorial pathophysiology that begins with the microcomedo as an initial step in comedogenesis and formation of subsequent acne lesions (Fig. 1).¹ Acne lesions can remain subclinical (microcomedones) or can continue to evolve into inflammatory lesions or comedones.¹

The follicular environment

Acne develops in the pilosebaceous unit with hyperseborrhea, and follicular obstruction is an early event in lesion initiation.² Obstruction can occur due to hyper-keratinization of the follicular epithelium.²⁻⁴ Cells accumulate until the pilosebaceous duct

is stretched to the maximum, then sometimes ruptures and leaks noxious stimuli into the surrounding tissue.

Inflammation

Jeremy *et al.* showed via immunohistochemistry that inflammatory markers are present in microcomedones.⁵ Specifically, IL-1 was expressed in early inflamed lesions (Fig. 2).⁵ Interestingly, IL-1 was found in comedonal contents both in the presence and absence of *P. acnes*; inflammatory components (CD4 T cells, macrophages, cytokines, and integrins) were also identified in the perifollicular area of uninvolved skin from acne patients.⁵

Dendritic cells pick up *P. acnes* antigenic material and T-cells primed at the local lymph node invade the follicular wall, leading to spongiosis which in turn allows influx and reflux of

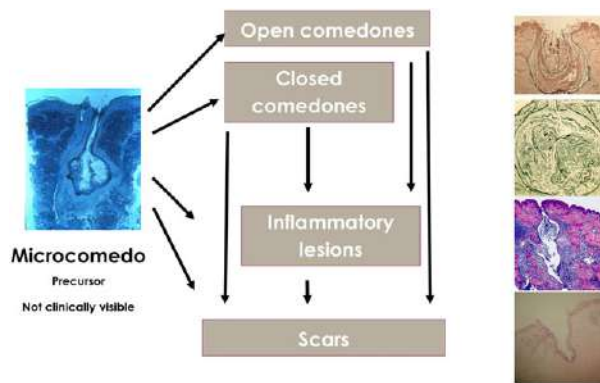


Figure 1 Lesion progression in acne. Adapted with permission from Gollnick et al. *J Am Acad Dermatol*. 2003.

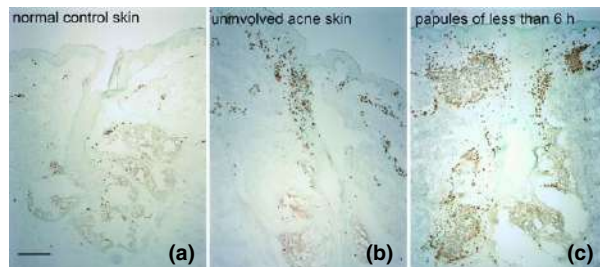


Figure 2 Inflammatory events are involved in the initiation of acne lesions. Reproduced with permission from Jeremy AHT, et al. *J Invest Dermatol*. 2003;121:20-27.

mediators.³ A recent hypothesis is that hypoxia due to increasing pressure from the ductal plug may serve as the link between a subclinical lesion and increased inflammatory responses at the pilosebaceous duct.³ Low oxygen levels allow *P. acnes* to grow in the infra-infundibulum and can stimulate keratinocytes to produce cytokines, which are released as the duct ruptures. Further, release of *P. acnes* into the dermis triggers an intense local reaction of the innate immune system.³ Collections of neutrophils and lymphocytes in the follicular epithelium also likely contribute to breakdown of the follicular wall (Fig. 2).⁵

Disturbance of the barrier function of the individual follicle could occur as a result of changes in sebum production that create a pro-inflammatory pattern, deficiency of linoleic acid, and production of pro-inflammatory cytokines by the sebocytes, leading to up-regulation of IL-1.⁶⁻¹⁰ IL-1, in turn, stimulates an inflammatory cascade and activation of local endothelial cells plus inflammation in the vasculature around the follicle (Fig. 3).¹¹ When occurring before hyper-proliferation of keratinocytes, some of these changes are similar to a type IV delayed hypersensitivity response.⁵ IL-1 as an initiator for hyper-keratinization has also been proposed, with the idea of 'keratinocyte activation cycles,' wherein activation of injured keratinocytes

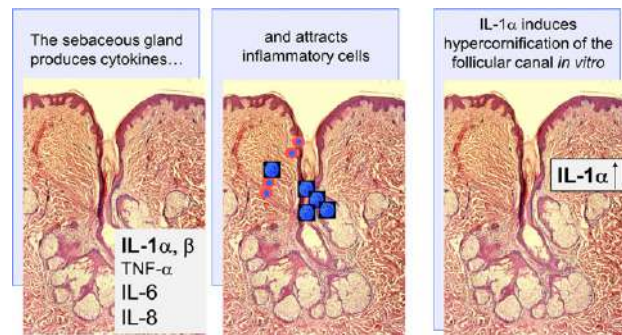


Figure 3 IL-1 triggers inflammation in the microcomedone to attract inflammatory cells and induce hypercornification of the follicular canal.

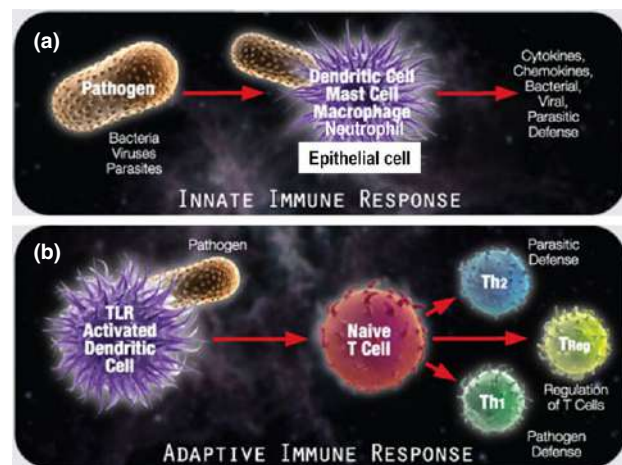


Figure 4 Innate and adaptive immunity. Modified with permission from Dranoff G. *Nature Rev Cancer*. 2004;4:11022.

leads to hyper-proliferation, migration, and selectin secretion, and also signals for migration of fibroblasts.^{12,13}

Although the exact sequences are not fully known, it seems possible that inflammatory events – namely production of IL-1 – occur early on and contribute to hyper-keratinization (Fig. 3).^{14,15} The importance of inflammation from the very beginning and throughout all stages of acne lesions is a relatively new concept.¹⁶

One digital photography study has suggested that inflammatory lesions can arise from both acne lesions and normal appearing skin (presumably via microcomedones).¹⁷ In this study, a subset of inflammatory lesions was not preceded by comedones, a finding that is consistent with the concept of sub-clinical inflammation as a primary event in acne.¹⁷ It is intriguing that some inflammatory lesions may come from clinically non-inflamed lesions (in addition to microcomedones); how-

Table 1 Components of the cutaneous innate immune system²⁰.

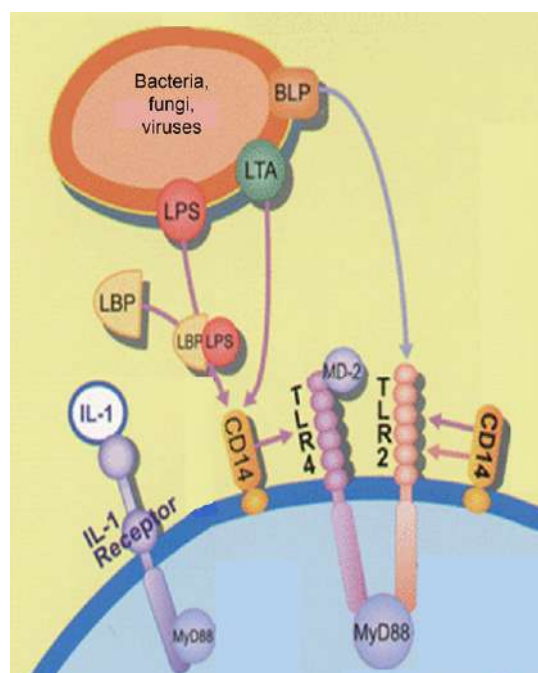
Skin cells
Interfollicular keratinocytes
Follicular keratinocytes
Merkel cells
Mast cells
Langerhans cells
Sebocytes
Soluble factors
Complement
Antimicrobial peptides
Chemokines
Cytokines
Metalloproteases, inhibitors of Metalloproteases
Hematopoietic cells
Monocytes/macrophages
Dendritic cells, plasmacytoid type
Natural killer (NK) cells
Polymorphonuclear cells (PMNs) or neutrophils

ever, the results of this initial study have not yet been reproduced. Slow degradation of non-viable follicular *P. acnes* contributes a long-standing proinflammatory stimulus, which can last through acne-related dyschromias (hyperpigmentation and erythema) as well as scarring.¹⁸ Inflammatory cell infiltrates have been found in acne scars; in a 2011 microscopy study, 77% of atrophic scars demonstrated inflammatory markers.¹⁹

The innate immune system is a lead actor in the protective function of skin

The human immune system has two major components: the nonspecific, innate immune system and the specific, adaptive immunity (Fig. 4). Innate immunity responses occur rapidly and provide the body's first line of defense against commonly shared pathogen structures. It protects the host in the early phase of an infectious challenge and relies on a set of germline-encoded receptors and molecules that recognize conserved molecular patterns found primarily in microorganisms. Several theories have attempted to explain the mechanisms involved in the regulation of lymphocyte activation in response to nonself-antigens. Most propose that the interaction of the pathogen-associated molecules with the innate response receptors/molecules results in activation of innate pathways which provide the signals required for the regulation of T- and B-cell activation and thus the adaptive immune system. The adaptive immune response requires time to target individual antigens; because of this investment, the adaptive immune system retains memory against any irritating or disease-causing agents that are encountered.^{20,21}

A primary function of skin is to recognize harmful versus non-harmful stimuli; in addition to being a physical barrier, there are many immune system elements in the skin that create

**Figure 5** Toll-like receptors in innate immunity.

an immunologic defense network resulting finally in a barrier such as in the intestine or respiratory tract. Elements of the immunological network in skin can be triggered by physical or chemical or mechanic injury or pathogens of different origin. There are multiple pathways designed to eliminate invading organisms as well as diseased or damaged host cells.²⁰ The primary components of the innate immunity in skin are cells of the skin, soluble factors, and different types of hematopoietic cells; specific examples of each are listed in Table 1.²⁰

The intact stratum corneum is the primary physical barrier; compromise of the stratum corneum via minor breaks (cuts, scratches), abnormalities secondary to disease, and other factors, greatly enhances the ability of microorganisms to cause infection. The stratum corneum, which is often conceptualized as a bricks and mortar, includes cross-linked cells with extracellular layers of lipids including ceramides, cholesterol, and free fatty acids.²⁰ The lipid layers are acidic due to the presence of free fatty acids; this provides protection by inhibiting growth of some bacteria (*Staphylococcus aureus* is one example). Soluble factors – complement, antimicrobial peptides, chemokines, and cytokines – are present in the epidermis and provide a chemical defense against invaders.²⁰ They are mainly produced by intra- and inter-follicular keratinocytes, sebocytes and the dendritic cells repertoire as well as by microbes themselves.

Keratinocytes, melanocytes and Langerhans cells in the epidermis can recruit dendritic cells, macrophages, natural killer cells and PMNs. In order to quickly recognize a large variety of

pathogenic organisms, these cells rely on well conserved pattern recognition receptors (PRRs), which are created by germline DNA and respond to particular pathogen-associated molecular patterns (PAMPs). PAMPs are shared by many pathogens, allowing the innate immune system to mount a fast response that is nonspecifically active against multiple challenges.²⁰

An important family of PRRs is known as toll-like receptors (TLRs); these receptors can initiate innate immune responses and later influence adaptive immune responses (Fig. 5).² TLRs are transmembrane proteins that respond to PAMPs such as bacterial cell wall components and genetic material. For example, TLR2 reacts to peptidoglycan from gram-positive bacteria while TLR4 is triggered by lipopolysaccharide from gram-negative bacteria.^{22,23} Activation of TLRs stimulates a variety of intracellular pathways that ultimately cause release of proinflammatory and immunomodulatory cytokines, including interleukins (ILs) and tumor necrosis factor alpha (TNF- α).^{20,24,25}

To summarize, any pathogen that manages to slip past the physical and chemical barriers of the epidermis are quickly recognized by pathogen receptors such as TLRs and then attacked by cells (PMNs or NKs) and/or secreted substances like antimicrobial peptides (AMPs). In the process, inflammatory responses can be triggered that lead to unwanted tissue self-injury, abnormal tissue growth cycles, and disturbances of down-regulation of immune responses in the local environment leading to disease.²

Evaluation of gene expression has shown that genes controlling a variety of innate immune system pathways are activated in acne patients.²⁶ It is thought that multiple innate immunity pathways contribute to acne pathogenesis, including alterations in barrier function, upregulation of inflammatory molecules and antimicrobial peptides, and activation of effector pathogen recognition pathways stimulated by *P. acnes*.^{1,2,24,27,28} Further,

several mechanisms link innate immunity with scarring, including an increase in MMPs or a modification of the ratio MMPs/TIMPs that promotes scar development and damage to dermal matrix and differences in immune response among patients who scar.^{29,30} *P. acnes* could play a role in scar formation due to enhanced extracellular matrix degradation by MMP-stimulated peptidoglycan from the bacterial cell wall.³⁰

Propionibacterium acnes is a trigger for innate immunity and promotes both acute and chronic inflammation

The gram-positive anaerobe *P. acnes* is a commensal organism commonly found on skin. It thrives in the milieu of the pilosebaceous canal and is found predominantly in sebaceous areas (face, chest, back). Its involvement in acne has been known for many years, despite that it also has been apparent that acne is not a traditional infectious disease with *P. acnes* as a pathogen.^{1,27} There are at least four primary pathways through which *P. acnes* interacts with the innate immune system to promote inflammation: via TLRs, by activating inflammasomes, by inducing production of matrix metalloproteinases (MMPs), and by stimulating activity of antimicrobial peptides. It should also be noted that different strains of *P. acnes* have varying effects on the innate immune system.^{11,31} Inflammation also occurs secondary to the adaptive response. Specific anti-*P. acnes* antibody can activate complement and attack neutrophils due to C5A production. Finally, as mentioned above, it has been postulated that there may be Type IV cellular response that also is adaptive.⁵

Interactions with TLRs

Propionibacterium acnes activates TLR-2 and TLR-4 on membranes of inflammatory cells.^{2,32} This leads to production of

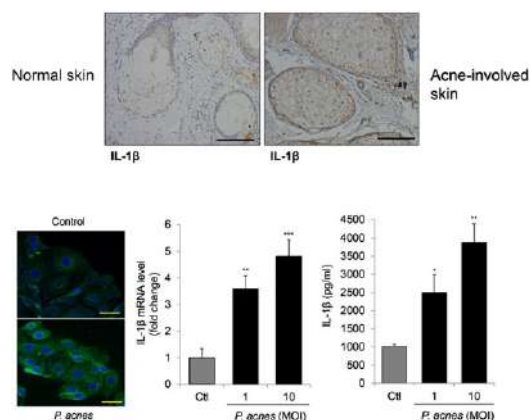


Figure 6 *Propionibacterium acnes* upregulates IL-1 β via activation of the NLRP3 inflammasome in human sebocytes.¹¹

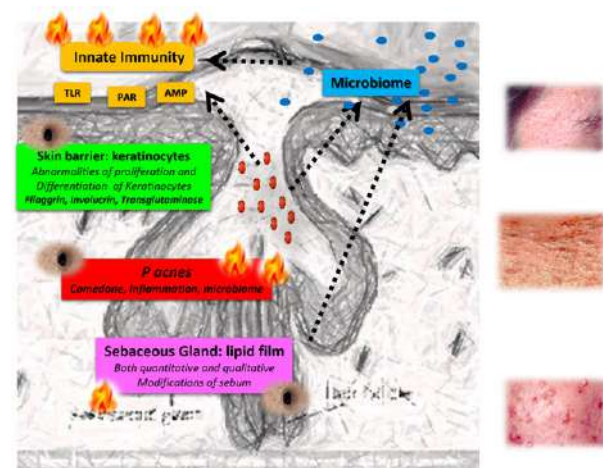


Figure 7 Schematic overview of interacting pathologic factors in acne.

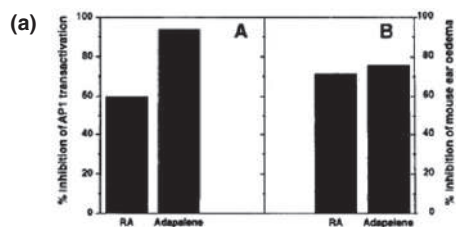


Figure 4. Inhibition by adapalene and RA of AP1 activity in vitro (A) and in vitro (B).

Br J Dermatol 1998;139 (suppl. 52), 3-7

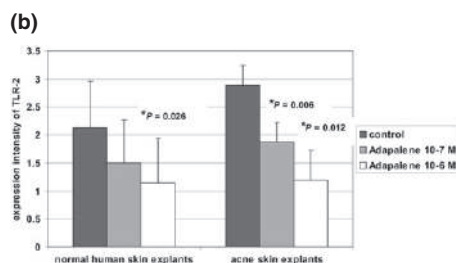


Figure 8 Inhibitory effect of topical retinoids on MMPs (a) and expression of TLRs (b). From Tenaud *et al.* with permission.⁴⁷

Table 2 Targeting innate immunity and inflammation in acne

	Innate response	immune	Clinical inflammation
Topical retinoids	+++		++
Systemic retinoids	+++		+++
Topical antibiotics	? (no data)		+
Systemic antibiotics (cyclines)	++		+++
BPO	++		++
Zinc	++		+

proinflammatory cytokines, including TNF- α and IL-1, IL-8, and IL-12.³³ In addition, it has been shown that more cells express TLR-2 as acne severity increases. This may be one explanation for why agents that target TLR-2, such as topical retinoids, have been shown to have greater efficacy in patients with more severe acne.³⁴ Cytokines are also produced in proportion to interaction between *P. acnes* and TLR-2, defensins and MMP via PAR-2R activation.³²

Inflammasomes and *P. acnes*

Propionibacterium acnes activates the inflammasome, a cytoplasmic complex of proteins that regulate cytokine activation and secretion (including IL-1).¹⁶ Inflammasome activation is accompanied by monocyte activation; the result is release of inflammatory mediators including IL-1 β .¹⁶ IL-1 is expressed as pro-proteins that require proteolytic processing; this processing

Table 3 Actions of oral tetracyclines in acne. Reprinted with permission from Monk *et al.*⁶³

Pathophysiological mechanisms in acne and corresponding non-antimicrobial actions of tetracyclines.

Acne	
Pathophysiology	Non-Antimicrobial Actions of Tetracyclines
Inflammation	• IL-8 inhibition • Inhibit neutrophil chemotaxis and activation • ROS inhibition • Inhibition of AA/5-LO pathway • MMP inhibition
Altered keratinization	• Inflammation inhibition • Cytokine inhibition

IL-8: interleukin-8; ROS: reactive oxygen species; MMP: matrix metalloproteinase; AA: arachidonic acid; 5-LO: 5-lipoxygenase.

Monk E *et al.* Pharm Res 63:130-45, 2011

is performed by caspase-1.¹⁶ Caspase gene expression also up-regulated and caspase-1 and NLRP3 (inflammasome gene) were found around pilosebaceous follicles, co-localized with tissue macrophages. Very recently, Qin *et al.* provided 'novel evidence that *P. acnes* contributes to the inflammatory response seen in acne by triggering the activation of the NLRP3-inflammasome in antigen-presenting cells, subsequently leading to enhanced IL-1 β secretion.³⁵ This group also reported that silencing NLRP3 attenuated the secretion of IL-1 induced by *P. acnes*. In 2014, Kistowska *et al.* reported that *P. acnes* triggers inflammation via a pathway driven by IL-1 β and the NLRP3 inflammasome of myeloid cells (Fig. 6).¹¹ They note that *P. acnes* can activate the NLRP3 inflammasome by a variety of mechanisms (generation of reactive oxygen species, potassium efflux, and lysosomal rupture). Further, bacterial phagocytosis has an essential role in NF- κ B dependent pro-IL1 β synthesis and inflammasome dependent cleavage into the activated form that is secreted. In summary, *P. acnes* is 'a potent trigger of NLRP3-inflammasome assembly and IL-1 β production ... [suggesting] sensing of *P. acnes* *in vivo* is a critical event in the exacerbation of inflammatory skin lesions.'¹¹ It is possible that in the future, specific inflammasome inhibitors may be developed as treatment for acne.¹⁶

Matrix metalloproteinases

Matrix metalloproteinases (MMPs) are involved in tissue destruction and scar formation and can participate in the innate immune response. In healthy skin, MMPs have an essential role in regulating the skin matrix.³⁰ *P. acnes* has been shown to upregulate several MMPs.^{36,37} The process is mediated by transcription factor activator protein-1 (AP-1) in acne lesions. Targeting MMPs has an important role in acne therapy as this may be a potential way to minimize scar development and abnormal skin remodeling.³⁸

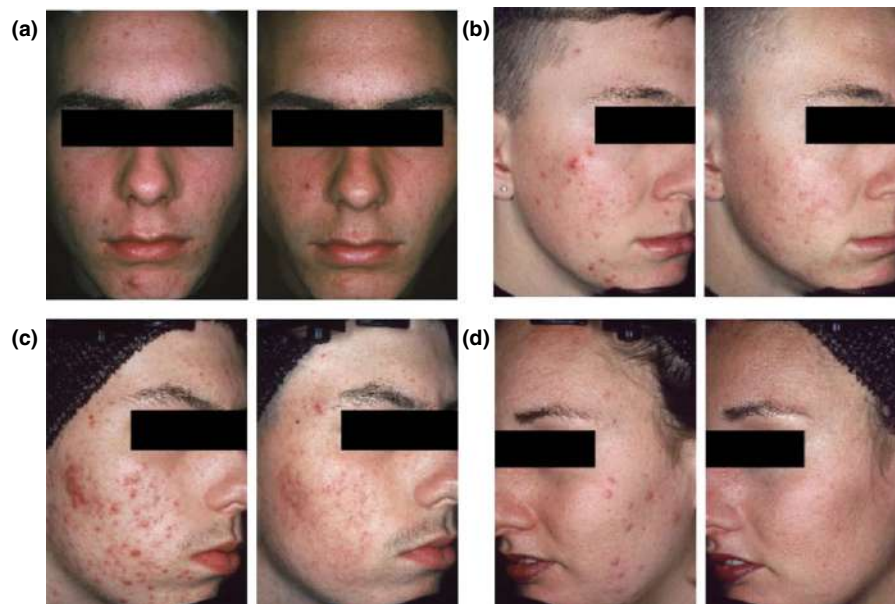


Figure 9 Effect of topical retinoid monotherapy on inflammatory acne. From Leyden *et al.* with permission.⁴⁸

Interactions with antimicrobial peptides (AMPs)

Another important point is that skin acts as a natural 'chemical barrier' via constitutively and inducibly produced AMPs, which include some β -defensins, RNase 7, the S100-protein psoriasin and cathelicidins. In acne, skin cells activate an antimicrobial response to *P. acnes* but in the process stimulate an excessive inflammatory response that exacerbates disease. Invading microorganisms face many challenges in the skin, not least of which are the antimicrobial peptides.^{39–41} The production of AMPs is mediated by the MyD88 pathway and IL-1 signaling. AMPs are bactericidal but also promote additional inflammatory responses. Lipids produced by the sebocyte exhibit antibacterial activity and AMPs have been found in human pilosebaceous units.³⁹ Sebum-free fatty acids may provide direct antibacterial activities against *P. acnes* by inducing the expression of human β -defensin-2.⁴² A very recent study suggests a direct link between the level of human β -defensin-2 in the skin and the development of papulopustular lesions.

Propionibacterium acnes and microbiome

There is also research into how the microbiome affects the impact of *P. acnes* on the skin. It has been shown that the quantity of *P. acnes* is not significantly different in the follicles of acne patients compared with controls.⁴³ This suggests that other factors besides sheer numbers play a role in how the skin reacts to *P. acnes*. This may be due to differing virulence among strains or to changes in virulence due to skin microbiome.⁴³ Acne has been theorized as a noninfectious disease influenced by shifts and imbalances in skin microbiome. As an

example, oral treatment with *Saccharomyces boulardii* was shown in controlled trials to influence acne via the intestinal microbiome. Reduction of sebum flow via anti-androgenic pills and oral isotretinoin (and in the future, perhaps topical drugs) gives *P. acnes* no chance to grow and build colonies. Further studies are needed to tease out the specific differences in *P. acnes* strains, host characteristics, and the microbiome characteristics.⁴³ A disturbance in the microbiome (or altered microbial balance) can trigger innate immunity and activate various inflammatory pathways.

A graphic summary of the interacting factors is presented in Fig. 7.

Clinical relevance: actions of current therapies on innate immune system and inflammation

The available anti-acne drug classes include retinoids (topical and systemic) and antimicrobials (antibiotics [topical/systemic], benzoyl peroxide, and zinc); hormonal therapy is also available but does not work through inflammatory pathways and will not be discussed further. Table 2 provides an overview of the effects of various anti-acne drug classes on innate immune response and inflammation.

Topical retinoids

Retinoids have multiple actions in acne, including well-known comedolytic and anti-comedogenic effects, targeting of the microcomedone, and anti-inflammatory actions (Table 3).⁴⁴ Topical retinoids affect/modulate multiple inflammatory pathways. These agents have been shown to inhibit leukocyte migration,

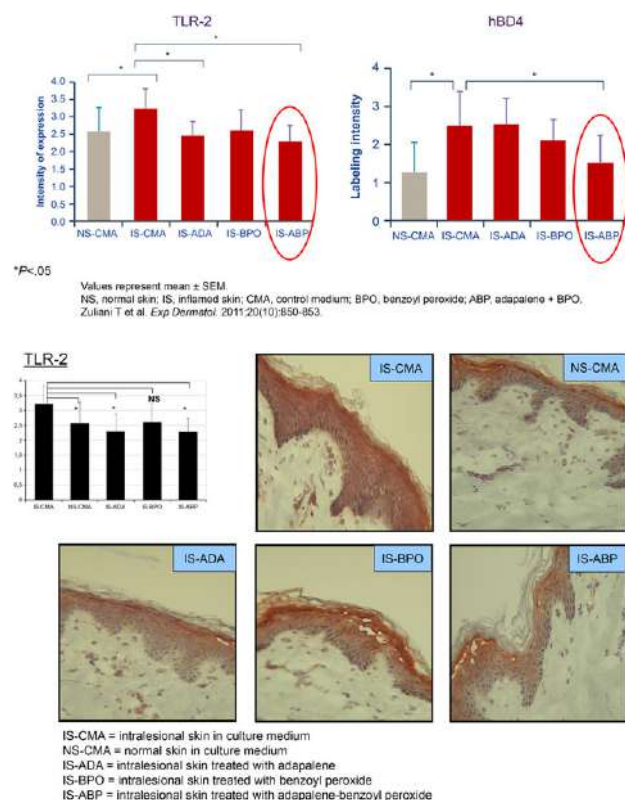


Figure 10 Synergistic effects of adapalene-BPO fixed dose combination. Reprinted with permission from Zuliani *et al.*⁵⁰

cytokine production, and arachidonic acid metabolism. Importantly, retinoids regulate the transcription factor AP-1.⁴⁴ This, in turn, results in inhibition of MMPs (Fig. 8), the enzymes responsible for scar formation in acne. *In vitro* studies suggest adapalene may have greater anti-inflammatory activity than other tested retinoids; although anti-inflammatory actions are a class effect of retinoids.^{45,46} Adapalene and tretinoin have been shown to down-regulate activation of TLRs (Fig. 8); these receptors facilitate recognition of microbial pathogens by monocytes and macrophages.⁴⁷

While there is a perception that topical retinoids act primarily on comedones, there is also strong clinical data showing the important efficacy of retinoids on inflammatory lesions.⁴⁸ Leyden *et al.* reported an investigator-blinded, vehicle controlled photographic assessment study of the efficacy of topical retinoids as monotherapy in inflammatory acne. This retrospective study evaluated the efficacy of retinoid alone in 577 patients who had participated in controlled clinical trials of tazarotene 0.1% gel, adapalene 0.1% gel, tretinoin 0.1% microsphere, tretinoin 0.025% gel, and tazarotene 0.1% cream. Fig. 9 shows clear clinical improvement, with greatest improvement occurring in more severe cases.⁴⁸

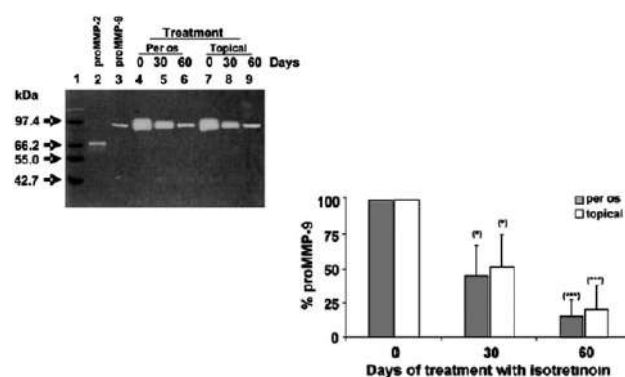


Figure 11 Isotretinoin inhibits tissue expression of upregulated MMPs in acne. Reprinted with permission from Papakonstantinou *et al.*⁶²

Retinoid fixed-dose combinations

The clinical efficacy of adapalene/BPO is greater than would be expected with a purely additive effect, suggesting that true synergy occurs between adapalene and BPO.⁴⁹ A synergistic effect can be defined as the cooperative action of two drugs resulting in a different or greater response than that of the individual drugs. Analysis of the clinical results with adapalene/BPO showed that the efficacy of the fixed-dose combination product is greater than the sum of the efficacy of adapalene alone plus BPO alone.⁴⁹ Combination therapy with adapalene/BPO has been shown to have synergistic effects on markers of inflammation and innate immune system activation, including TLRs.⁵⁰ Zuliani *et al.* studied the modulatory effects of adapalene and BPO alone or in the fixed-dose combination on several parameters: keratinocyte expression of non-specific immunity receptors, keratinocyte proliferation/differentiation markers, metalloproteinases, pro/anti-inflammatory cytokines, and the TACE enzyme (Fig. 10). The actions of the molecules were examined in normal skin explants and biopsies from papules on the back of acne patients. A synergistic effect was exerted on markers of keratinocyte proliferation/differentiation (Ki67, transglutaminase, and $\alpha 2$ and $\alpha 6$ integrins). Synergistic actions of adapalene/BPO on aspects of innate immunity was also observed, and included decreasing toll-like receptor 2 (TLR2), matrix metalloproteinase 3 (MMP3), interleukin 8 (IL-8) and interleukin 10 (IL-10). In addition, adapalene/BPO had a synergistic effect in inducing $\beta 4$ defensin. Thus, the synergistic effect seen at the clinical level has also been demonstrated at the molecular level.⁵⁰

Isotretinoin

Isotretinoin has been shown to normalize the response of the innate immune system to *P. acnes*; however, it should be noted that it is unclear whether this is a direct effect or secondary to reduction of new lesion formation.⁵¹ Dispenza *et al.* reported that this was due to a significant decrease in TLR-2 expression

by monocytes, with a subsequently lower cytokine response. Further, the effect was sustained for 6 months after isotretinoin therapy was discontinued, suggesting that this may be responsible for the long-lasting therapeutic effects of isotretinoin.⁵¹ Papakonstantinou *et al.* reported that isotretinoin normalizes the follicular environment by inhibiting MMPs (Fig. 11). Oral isotretinoin also has anti-androgenic effects.⁵²

Antibiotics

Oral antibiotics as cyclines at low doses may affect *P. acnes* levels but are thought to have a primarily anti-inflammatory effects. They inhibit the chemotaxis of granulocytes, inhibit the formation of granuloma, and limit the production of pro-inflammatory cytokines and metalloproteases. Notably, subantimicrobial doses of tetracyclines show these anti-inflammatory effects, perhaps due to inhibition of protein synthesis.^{53–55} Cyclines also work through oxidative pathways by decreasing activity of neutrophil-derived reactive oxygen species (ROS). This is an important pathway in acne, since patients with inflammatory acne have abnormally elevated levels of ROS compared to individuals with healthy skin or comedonal acne. Clindamycin has been shown to have some effects on follicular ceramides which may contribute to a mild anti-comedogenic effect.⁵⁶

Benzoyl peroxide (BPO)

A potent oxidizing agent, BPO is bactericidal by generating reactive oxygen species; microbial resistance against BPO has not been shown since its introduction. In addition to being bactericidal, BPO also has some dose-dependent comedolytic (but not anti-comedogenic) and anti-inflammatory effects.^{27,57} Acne treatment guidelines generally do not recommend use of BPO alone but in combination with a topical retinoid.^{58–60}

Conclusions

Acne is a multifactorial chronic inflammatory disease of the pilosebaceous unit. Early subclinical inflammation is a primary event and in association with increased sebum flow and hyperkeratinization of the follicular cells followed by microcomedo formation and by the increasing hosting of *P. acnes*. Microcomedones then can evolve to either clinically inflamed lesions (papules, pustules, nodules) or comedones. Inflammation is now also known to be persistent throughout the acne life cycle. Some anti-acne drugs, including retinoids, regulate immune system responses; today the treatment targets should be reconsidered with an emphasis on multiple pathophysiologic factors, including multiple innate immune system and anti-inflammatory pathways.⁵⁷ For the majority of patients, appropriate first-line acne management includes a combination of a topical retinoid plus an antimicrobial agent.^{1,27} This combination allows targeting of inflammation and the innate immune system via multiple pathways and has been shown to provide faster and superior results compared to antibiotic/antimicrobial

therapy alone.^{1,27} Further, global concerns about preservation of antibiotic utility for public health reasons necessitate minimizing reliance on this class of drugs in acne, where they are often used for long periods of time.⁶¹

References

- Gollnick H, Cunliffe W, Berson D *et al.* Management of acne: a report from a Global Alliance to Improve Outcomes in Acne. *J Am Acad Dermatol* 2003; **49**: S1–S37.
- Kim J. Review of the innate immune response in acne vulgaris: activation of Toll-like receptor 2 in acne triggers inflammatory cytokine responses. *Dermatology* 2005; **211**: 193–198.
- Danby FW. Ductal hypoxia in acne: is it the missing link between comedogenesis and inflammation? *J Am Acad Dermatol* 2014; **70**: 948–949.
- Kurokawa I, Mayer-da-Silva A, Gollnick H, Orfanos CE. Monoclonal antibody labeling for cytokeratins and filaggrin in the human pilosebaceous unit of normal, seborrheic and acne skin. *J Invest Dermatol* 1988; **91**: 566–571.
- Jeremy AH, Holland DB, Roberts SG, Thomson KF, Cunliffe WJ. Inflammatory events are involved in acne lesion initiation. *J Invest Dermatol* 2003; **121**: 20–27.
- Elias PM, Brown BE, Ziboh VA. The permeability barrier in essential fatty acid deficiency: evidence for a direct role for linoleic acid in barrier function. *J Invest Dermatol* 1980; **74**: 230–233.
- Perisho K, Wertz PW, Madison KC, Stewart ME, Downing DT. Fatty acids of acylceramides from comedones and from the skin surface of acne patients and control subjects. *J Invest Dermatol* 1988; **90**: 350–353.
- Zouboulis CC, Jourdan E, Picardo M. Acne is an inflammatory disease and alterations of sebum composition initiate acne lesions. *J Eur Acad Dermatol Venereol* 2014; **28**: 527–532.
- Picardo M, Ottaviani M, Camera E, Mastrofrancesco A. Sebaceous gland lipids. *Dermatoendocrinol* 2009; **1**: 68–71.
- Wertz PW, Swartzendruber DC, Abraham W, Madison KC, Downing DT. Essential fatty acids and epidermal integrity. *Arch Dermatol* 1987; **123**: 1381–1384.
- Kistowska M, Gehrke S, Jankovic D *et al.* IL-1 β drives inflammatory responses to propionibacterium *acnes* *in vitro* and *in vivo*. *J Invest Dermatol* 2014; **134**: 677–685.
- Freedberg IM, Tomic-Canic M, Komine M, Blumenberg M. Keratins and the keratinocyte activation cycle. *J Invest Dermatol* 2001; **116**: 633–640.
- Latkowski JM, Freedberg IM, Blumenberg M. Keratinocyte growth factor and keratin gene regulation. *J Dermatol Sci* 1995; **9**: 36–44.
- Guy R, Kealey T. Modelling the infundibulum in acne. *Dermatology* 1998; **196**: 32–37.
- Guy R, Kealey T. The effects of inflammatory cytokines on the isolated human sebaceous infundibulum. *J Invest Dermatol* 1998; **110**: 410–415.
- Contassot E, Beer HD, French LE. Interleukin-1, inflammasomes, autoinflammation and the skin. *Swiss Med Wkly* 2012; **142**: w13590.
- Do TT, Zarkhin S, Orringer JS *et al.* Computer-assisted alignment and tracking of acne lesions indicate that most inflammatory lesions arise from comedones and *de novo*. *J Am Acad Dermatol* 2008; **58**: 603–608.
- Vowels BR, Yang S, Leyden JJ. Induction of proinflammatory cytokines by a soluble factor of Propionibacterium *acnes*: implications for chronic inflammatory acne. *Infect Immun* 1995; **63**: 3158–3165.
- Lee WJ, Jung HJ, Lim HJ, Jang YH, Lee SJ, Kim DW. Serial sections of atrophic acne scars help in the interpretation of microscopic findings and the selection of good therapeutic modalities. *J Eur Acad Dermatol Venereol* 2013; **27**: 643–646.
- Fitzpatrick's Dermatology in General Medicine. New York: McGraw Hill; 2008.

- 21 Gardy JL, Lynn DJ, Brinkman FS, Hancock RE. Enabling a systems biology approach to immunology: focus on innate immunity. *Trends Immunol* 2009; **30**: 249–262.
- 22 Yoshimura A, Lien E, Ingalls RR, Tuomanen E, Dziarski R, Golenbock D. Cutting edge: recognition of Gram-positive bacterial cell wall components by the innate immune system occurs via Toll-like receptor 2. *J Immunol* 1999; **163**: 1–5.
- 23 Poltorak A, He X, Smirnova I *et al.* Defective LPS signaling in C3H/HeJ and C57BL/10ScCr mice: mutations in Tlr4 gene. *Science* 1998; **282**: 2085–2088.
- 24 McInturf JE, Kim J. The role of toll-like receptors in the pathophysiology of acne. *Semin Cutan Med Surg* 2005; **24**: 73–78.
- 25 McInturf JE, Modlin RL, Kim J. The role of toll-like receptors in the pathogenesis and treatment of dermatological disease. *J Invest Dermatol* 2005; **125**: 1–8.
- 26 Trivedi NR, Gilliland KL, Zhao W, Liu W, Thiboutot DM. Gene array expression profiling in acne lesions reveals marked upregulation of genes involved in inflammation and matrix remodeling. *J Invest Dermatol* 2006; **126**: 1071–1079.
- 27 Thiboutot D, Gollnick H, Bettoli V *et al.* New insights into the management of acne: an update from the Global Alliance to Improve Outcomes in Acne group. *J Am Acad Dermatol* 2009; **60**: S1–S50.
- 28 Harper JC, Thiboutot DM. Pathogenesis of acne: recent research advances. *Adv Dermatol* 2003; **19**: 1–10.
- 29 Holland DB, Jeremy AH, Roberts SG, Seukeran DC, Layton AM, Cunliffe WJ. Inflammation in acne scarring: a comparison of the responses in lesions from patients prone and not prone to scar. *Br J Dermatol* 2004; **150**: 72–81.
- 30 Sato T, Kurihara H, Akimoto N, Noguchi N, Sasatsu M, Ito A. Augmentation of gene expression and production of promatrix metalloproteinase 2 by *Propionibacterium acnes*-derived factors in hamster sebocytes and dermal fibroblasts: a possible mechanism for acne scarring. *Biol Pharm Bull* 2011; **34**: 295–299.
- 31 Jasson F, Nagy I, Knol AC, Zuliani T, Khammari A, Dreno B. Different strains of *Propionibacterium acnes* modulate differently the cutaneous innate immunity. *Exp Dermatol* 2013; **22**: 587–592.
- 32 Jugeau S, Tenaud I, Knol AC *et al.* Induction of toll-like receptors by *Propionibacterium acnes*. *Br J Dermatol* 2005; **153**: 1105–1113.
- 33 Kim J, Ochoa MT, Krutzik SR *et al.* Activation of toll-like receptor 2 in acne triggers inflammatory cytokine responses. *J Immunol* 2002; **169**: 1535–1541.
- 34 Feldman SR, Tan J, Poulin Y, Dirschka T, Kerrouche N, Manna V. The efficacy of adapalene-benzoyl peroxide combination increases with number of acne lesions. *J Am Acad Dermatol* 2011; **64**: 1085–1091.
- 35 Qin M, Pirouz A, Kim MH, Krutzik SR, Garban HJ, Kim J. *Propionibacterium acnes* Induces IL-1 β secretion via the NLRP3 inflammasome in human monocytes. *J Invest Dermatol* 2014; **134**: 381–388.
- 36 Jalian HR, Liu PT, Kanchanapoomi M, Phan JN, Legaspi AJ, Kim J. All-trans retinoic acid shifts *Propionibacterium acnes*-induced matrix degradation expression profile toward matrix preservation in human monocytes. *J Invest Dermatol* 2008; **128**: 2777–2782.
- 37 Choi JY, Piao MS, Lee JB, Oh JS, Kim IG, Lee SC. *Propionibacterium acnes* stimulates pro-matrix metalloproteinase-2 expression through tumor necrosis factor- α in human dermal fibroblasts. *J Invest Dermatol* 2008; **128**: 846–854.
- 38 Kang S, Cho S, Chung JH, Hammerberg C, Fisher GJ, Voorhees JJ. Inflammation and extracellular matrix degradation mediated by activated transcription factors nuclear factor- κ B and activator protein-1 in inflammatory acne lesions *in vivo*. *Am J Pathol* 2005; **166**: 1691–1699.
- 39 Chronnell CM, Ghali LR, Ali RS *et al.* Human beta defensin-1 and -2 expression in human pilosebaceous units: upregulation in acne vulgaris lesions. *J Invest Dermatol* 2001; **117**: 1120–1125.
- 40 Lee DY, Yamasaki K, Rudisil J *et al.* Sebocytes express functional cathelicidin antimicrobial peptides and can act to kill *propionibacterium acnes*. *J Invest Dermatol* 2008; **128**: 1863–1866.
- 41 Ganceviciene R, Fimmel S, Glass E, Zouboulis CC. Psoriasin and follicular hyperkeratinization in acne comedones. *Dermatology* 2006; **213**: 270–272.
- 42 Nakatsui T, Kao MC, Zhang L, Zouboulis CC, Gallo RL, Huang CM. Sebum free fatty acids enhance the innate immune defense of human sebocytes by upregulating beta-defensin-2 expression. *J Invest Dermatol* 2010; **130**: 985–994.
- 43 Fitz-Gibbon S, Tomida S, Chiu BH *et al.* *Propionibacterium acnes* strain populations in the human skin microbiome associated with acne. *J Invest Dermatol* 2013; **133**: 2152–2160.
- 44 Czernielewski J, Michel S, Bouclier M, Baker M, Hensby JC. Adapalene biochemistry and the evolution of a new topical retinoid for treatment of acne. *J Eur Acad Dermatol Venereol* 2001; **15**(Suppl 3): 5–12.
- 45 Allec J, Chatelus A, Wagner N. Skin distribution and pharmaceutical aspects of adapalene gel. *J Am Acad Dermatol* 1997; **36**: S119–S125.
- 46 Shroot B, Michel S. Pharmacology and chemistry of adapalene. *J Am Acad Dermatol* 1997; **36**: S96–S103.
- 47 Tenaud I, Khammari A, Dreno B. *In vitro* modulation of TLR-2, CD1d and IL-10 by adapalene on normal human skin and acne inflammatory lesions. *Exp Dermatol* 2007; **16**: 500–506.
- 48 Leyden JJ, Shalita A, Thiboutot D, Washenik K, Webster G. Topical retinoids in inflammatory acne: a retrospective, investigator-blinded, vehicle-controlled, photographic assessment. *Clin Ther* 2005; **27**: 216–224.
- 49 Tan J, Gollnick HP, Loesche C, Ma YM, Gold LS. Synergistic efficacy of adapalene 0.1%-benzoyl peroxide 2.5% in the treatment of 3855 acne vulgaris patients. *J Dermatolog Treat* 2011; **22**: 197–205.
- 50 Zuliani T, Khammari A, Chaussy H, Knol AC, Dreno B. *Ex vivo* demonstration of a synergistic effect of Adapalene and benzoyl peroxide on inflammatory acne lesions. *Exp Dermatol* 2011; **20**: 850–853.
- 51 Dispenza MC, Wolpert EB, Gilliland KL *et al.* Systemic isotretinoin therapy normalizes exaggerated TLR-2-mediated innate immune responses in acne patients. *J Invest Dermatol* 2012; **132**: 2198–2205.
- 52 Mobacken H, Sundstrom A, Vahlquist A. 30 years with isotretinoin. 'Miracle medicine' against acne with many side effects. *Läkartidningen* 2014; **111**: 93–96.
- 53 Jain A, Sangal L, Basal E, Kaushal GP, Agarwal SK. Anti-inflammatory effects of erythromycin and tetracycline on *Propionibacterium acnes* induced production of chemotactic factors and reactive oxygen species by human neutrophils. *Dermatol Online J* 2002; **8**: 2.
- 54 Webster GF. Inflammation in acne vulgaris. *J Am Acad Dermatol* 1995; **33**: 247–253.
- 55 Akamatsu H, Kurokawa I, Nishijima S, Asada Y. Inhibition of neutrophil chemotactic factor production in comedonal bacteria by subminimal inhibitory concentrations of erythromycin. *Dermatology* 1992; **185**: 41–43.
- 56 Thielitz A, Helmdach M, Ropke EM, Gollnick H. Lipid analysis of follicular casts from cyanoacrylate strips as a new method for studying therapeutic effects of antiacne agents. *Br J Dermatol* 2001; **145**: 19–27.
- 57 Kirck LH. Re-evaluating treatment targets in acne vulgaris: adapting to a new understanding of pathophysiology. *J Drugs Dermatol* 2014; **13**: s57–s60.
- 58 Gold LS, Tan J, Cruz-Santana A *et al.* A North American study of adapalene-benzoyl peroxide combination gel in the treatment of acne. *Cutis* 2009; **84**: 110–116.
- 59 Dutil M. Benzoyl peroxide: enhancing antibiotic efficacy in acne management. *Skin Therapy Lett* 2010; **15**: 5–7.
- 60 Pariser D, Bucko A, Fried R *et al.* Tretinoin gel microsphere pump 0.04% plus 5% benzoyl peroxide wash for treatment of acne vulgaris: morning/morning regimen is as effective and safe as morning/evening regimen. *J Drugs Dermatol* 2010; **9**: 805–813.
- 61 Thiboutot D, Dreno B, Gollnick H *et al.* A call to limit antibiotic use in acne. *J Drugs Dermatol* 2013; **12**: 1331–1332.
- 62 Papakonstantinou E, Aletras AJ, Glass E *et al.* Matrix metalloproteinases of epithelial origin in facial sebum of patients with acne and their regulation by isotretinoin. *J Invest Dermatol* 2005; **125**: 673–684.
- 63 Monk E, Shalita A, Siegel DM. Clinical applications of non-antimicrobial tetracyclines in dermatology. *Pharmacol Res* 2011; **63**: 130–145.